COMP3016 Coursework 1 – Game Proposal

Game title: ProtoLaunch

Genre: Arcade/Endless Run

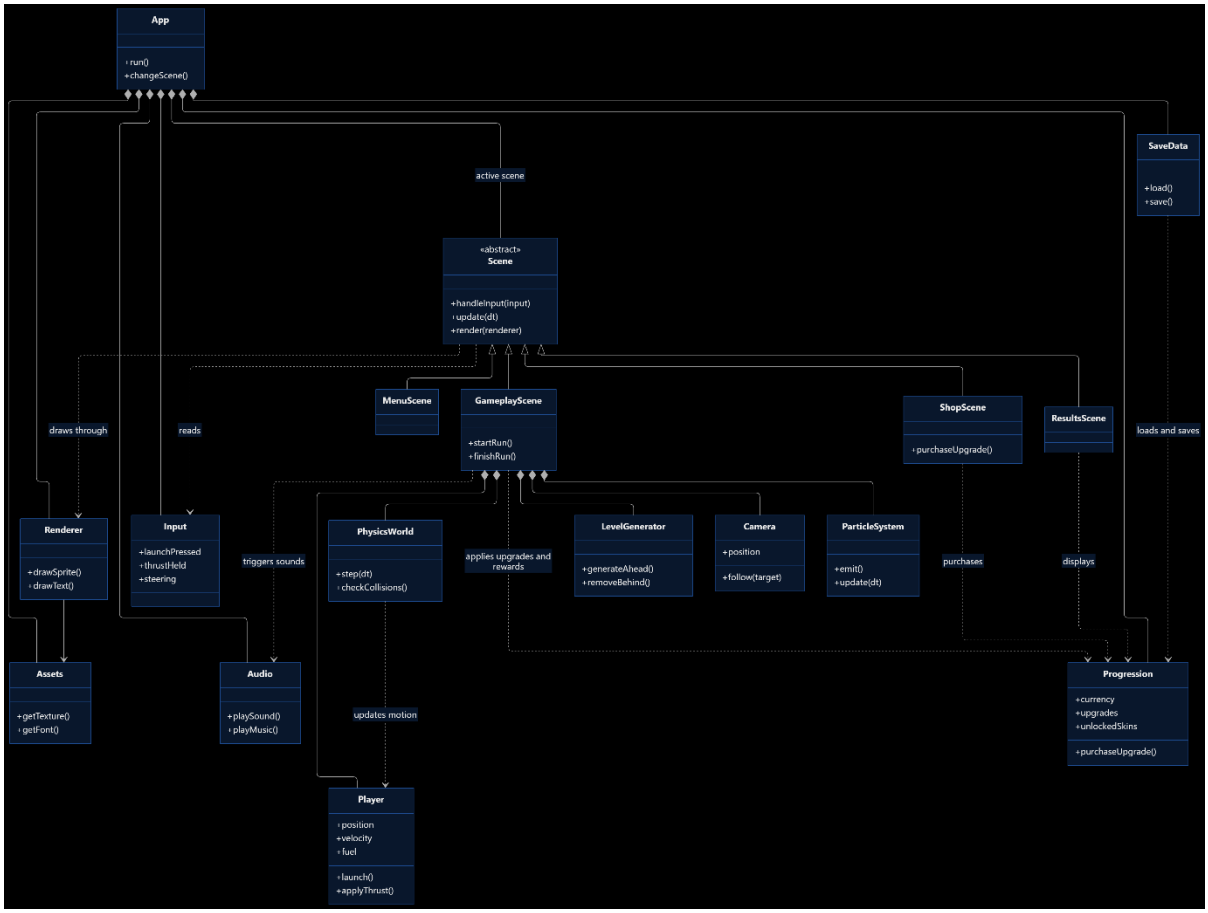
Synopsis: The player pulls back on an elasticated launcher, launching an object into the air. The aim is to collect as many coins as possible, bounce on as many trampolines as possible and reach the greatest distance. This will be achieved via the use of coins to purchase upgrades to progress further.

Core gameplay loop: Launch -> collect -> receive -> upgrade -> launch, with the end goal being achieve the highest score of distance + height.

Intended mechanics: Click and hold left mouse to pull back on the launcher to choose power and angle using the mouse. Release left mouse button to launch object.

Intended technical features: Physics, collision, randomly generated map

UML Sketch:



Architecture overview:

Layer	Responsibilities
Application	Initialise SDL and libraries, run the main loop, handle pause/resume and shutdown
Scenes	Main menu, gameplay, upgrade shop, and results screen
Game simulation	Player movement, physics, collisions, pickups, level generation, and run progress
Rendering	Draw backgrounds, sprites, particles, camera effects, and UI
Services	Load assets, play audio, read input, and save progression

External libraries: Box2D

Intended use of AI: Sprite generation, UI assets, optimisation, balancing, some code generation.

Development roadmap:

Phase	Work	Completion milestone
1. Project foundation	**Set up C++, CMake, SDL3, image loading, audio, and fonts. Add the main loop, input handling, and scene switching.**	**A window displays a sprite and text, responds to input, and plays a sound.**
2. First playable launch	**Implement the player, launch controls, gravity, camera tracking, and landing detection. Use a fixed physics timestep.**	**You can launch, control the ball, land, and restart.**
3. Complete the gameplay loop	**Add pickups, obstacles, distance/height tracking, rewards, and the results screen. Implement level generation and removal of distant objects.**	**Launch -> collect -> finish -> receive rewards -> launch again works.**
4. Progression and saving	**Add the upgrade shop, currency, unlockable skins, balancing data, and save/load support.**	**Progress survives restarting the application, and upgrades affect subsequent runs.**
5. Visual and audio polish	**Integrate backgrounds and sprite sheets. Add animation, particles, trails, screen shake, music, and sound feedback.**	**One complete level represents the intended final quality.**